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In the past two centuries, human activities have caused major changes in the tree species composition of southern Québec, Canada¹

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Abstract: We compared the presettlement forest composition of 60 lots of southern Québec, as reconstructed from wood sales reported in old notary deeds, with their present day composition. Some tree species, such as sugar maple (*Acer saccharum*) and ash (*Fraxinus americana* and/or *F. pennsylvanica*), have shown important increases since European colonization, while others have decreased sharply. In a site-by-site comparison, there are many instances where a species sold from a particular lot is completely absent from this lot today. Tamarack (*Larix laricina*) and black spruce (*Picea mariana*), two species that appear in the notary deeds, were completely eliminated from the studied lots. The apparent discrepancy between the predominance of lowlands in the landscape and the low presence in the notary deeds of tree species that prefer more humid conditions suggests that precolonial forests on the clay plain and lowlands were sparse or patchy or were composed of tree species of little stature or commercial value. For mesic sites, sales reported in the notary deeds suggest that the presettlement forest in the Haut-Saint-Laurent region corresponded to the maple-beech-birch forest type and that the most important change since European settlement is a loss or decrease in the abundance of American beech (*Fagus grandifolia*) and yellow birch (*Betula alleghaniensis*). In this agro-forested landscape, there is no indication of an eventual return toward the original forest composition, nor should we expect one considering the extent and the irreversible nature of some of the ecosystem modifications caused by human activity.

Keywords: presettlement forests, succession, American beech, sugar maple, yellow birch.

Résumé : Nous avons comparé la composition forestière précoloniale de 60 lots du Sud du Québec, reconstituée à partir de ventes de bois consignées dans des actes notariés, à leur composition actuelle. Certaines espèces comme l'Érable à sucre (*Acer saccharum*) et les frênes (*Fraxinus americana* et/ou *F. pennsylvanica*) montrent une hausse importante dans leur abondance depuis la colonisation européenne, alors que d'autres ont connu une décroissance substantielle. Nous avons noté à plusieurs reprises qu'une espèce autrefois récoltée et vendue sur un lot est aujourd'hui complètement absente de ce même lot sous sa forme arborescente. Le Mélèze laricin (*Larix laricina*) et l'Épinette noire (*Picea mariana*), deux espèces qui apparaissent dans les actes notariés, ont complètement été éliminées des lots étudiés. L'apparente contradiction entre la prédominance des basses terres dans le paysage et la faible présence d'espèces qui préfèrent les conditions humides dans les actes notariés suggère que les forêts précoloniales de la plaine d'argile et des basses terres étaient peu denses, disparates ou dominées par des espèces forestières de petite taille ou de faible valeur commerciale. Chez les sites mésiques, les résultats suggèrent que la forêt précoloniale du Haut-Saint-Laurent correspond à une érablière à hêtre (*Fagus grandifolia*) et à Bouleau jaune (*Betula alleghaniensis*) et que le changement le plus important depuis la colonisation européenne est une décroissance dans l'abondance du hêtre et du Bouleau jaune. Dans ce paysage agroforestier, il ne semble pas y avoir de tendance vers un retour éventuel de la composition forestière originale, d'autant plus si l'on considère la sévérité et la nature irréversible de certaines modifications de l'écosystème causées par l'activité humaine.

Mots-clés : forêt précoloniale, succession, hêtre, Érable à sucre, Bouleau jaune.

Nomenclature: Marie-Victorin, 1997.

Introduction

The reconstruction of presettlement forest characteristics, by providing a baseline standard for evaluating present-day environments, is considered an essential component of conservation planning and forest management (Foster, 1992; Frelich, 1995; Cogbill, 2000; Jackson *et al.*, 2000). Witness trees from early survey records have been widely used to assess forest change since European settlement in eastern North America (Manies & Mladenoff,

2000; Dyer, 2001; Schulte & Mladenoff, 2001). Studies based on survey records have shown alteration in forest structure and composition due to historical land-use practices, providing a better understanding of the human influence on the forest landscape. For example, there was a significant shift in forest composition and structure in southeastern Ohio, but the relative abundance of tree species in the present-day landscape is similar to that in the presettlement forest (Dyer, 2000). Siccama (1971) found no dramatic changes in the presence or absence of tree species in northern Vermont, but did report "an overall increase in the importance of pine, hardshack and poplar in the present forests on the intermediate elevations

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which reflects the stages of secondary succession on much abandoned farmland". Conversely, Cogbill (2000) noted that the substantial decline of spruce at mid-elevation caused by climatic changes and forest harvesting in northern New England was nearly balanced by substantial gains in the valleys due to regeneration in old fields.

Located near the northern limit of the deciduous forest of North America, the Haut-Saint-Laurent region of Québec remained untouched and nearly uninhabited until the end of the 18th century. The need for agricultural lands and the intensive logging activities of the early 19th century considerably changed the original landscape. Since European colonization, a variety of disturbances have created a complex agro-forested mosaic. In this region, as in the rest of eastern North America, there is a growing recognition that managing present-day forest stands to achieve specific objectives (*i.e.*, wood production, conservation, etc.) should be based on a better understanding of the nature and consequences of past transformations (Bouchard & Domon, 1997). Early survey records in Québec occasionally included brief qualitative descriptions of forest composition. However, the exact reconstruction of the presettlement forest from these records is not possible because survey methods did not use witness trees as in the United States. An alternative approach based on wood sales reported in early notary deeds was developed to reconstruct past forest composition (Bouchard *et al.*, 1989; Simard & Bouchard, 1996). Produced from the very beginning of the European settlement in the region, these deeds generally included the quantity of wood sold, the type (species), and the price of the transaction. Changes in the relative abundance of tree species sold over the early period, with consideration of their evolving commercial value, provide an indication of changes in forest composition. For example, white pine (*Pinus strobus*) wood was frequently sold early in the 19th century, until near depletion in the 1840s. Bur oak (*Quercus macrocarpa*), although much less abundant than pine, was also harvested very early due to its high value. American beech (*Fagus grandifolia*) and yellow birch (*Betula alleghaniensis*), two species with low abundance in present-day forests of the Haut-Saint-Laurent region, were frequently sold between 1810 and 1850 (Bouchard *et al.*, 1989; Simard & Bouchard, 1996). Because the goal of these studies was to examine broad-scale changes in sales patterns over time for the Haut-Saint-Laurent region as a whole, no attempt was made to determine the exact spatial origin of the wood harvested.

While the location of the harvest is not reported in the notary deeds, it is often possible to infer the origin of the wood harvested by assigning land ownership to the seller using other archival documents. In the present study, we make a spatially explicit comparison of presettlement forests with present-day composition by relating wood sales reported in the notary deeds to the location of their harvest. The results of our site-by-site comparison show major changes in tree species composition, with many species completely absent from sites where they were once harvested. Conclusions from previous studies and inferences from historical documents are re-examined in the light of our new findings.

Methods

STUDY SITES

The studied lots are located within or immediately adjacent to the township of St-Anicet, in the Haut-Saint-Laurent region of southern Québec (Figure 1). St-Anicet Township is bordered to the northwest by Lake St-François, which is a widening of the St. Lawrence River, and by the townships of Dundee, Godmanchester, and Ste-Barbe. The region has a humid continental climate with hot summers and cold winters. The mean annual temperature and total precipitation are 6.5°C and 1,006.9 mm for the years 1961 to 1990 (Huntingdon meteorological station). The mean temperature reaches a high of 21°C in July and a minimum of -9.5°C in January (Environment Canada, 1993). The bedrock is composed of Beekmantown dolomite, which is the main element found in regional moraine deposits. The relief is mostly flat, with ridges and mounds of morainic deposits emerging from the marine plain of clay deposits (Bariteau, 1988). A sandy area of littoral origin is located in the western part of the study area, and a large peatland mostly located outside the study area makes a small incursion in the southeast.

The Haut-Saint-Laurent region is located within the Maple-Bitternut Hickory Bioclimatic Domain (Ministère des Ressources naturelles du Québec, 2003) and the St. Lawrence Lowlands Ecoregion (Ecoregions Working Group, 1989). The Haut-Saint-Laurent has been the subject of several recent studies on its vegetation (Bergeron, Bouchard & Leduc, 1988; Brisson, Bergeron & Bouchard, 1988; Meilleur, Bouchard & Bergeron, 1992; Jean & Bouchard, 1987, 1991, 1993; Meilleur, Bouchard & Bergeron, 1994; de Blois & Bouchard, 1995; Doyon, Bouchard & Gagnon, 1998; Bouchard & Jean, 2001; de Blois, Domon & Bouchard, 2001) and land-use dynamics (Domon, Bouchard & Gariépy, 1993; Paquette & Domon, 1997; Pan *et al.*, 1999, 2001). Mesic forests are generally dominated by sugar maple (*Acer saccharum*), with basswood (*Tilia americana*), ironwood (*Ostrya virginiana*),

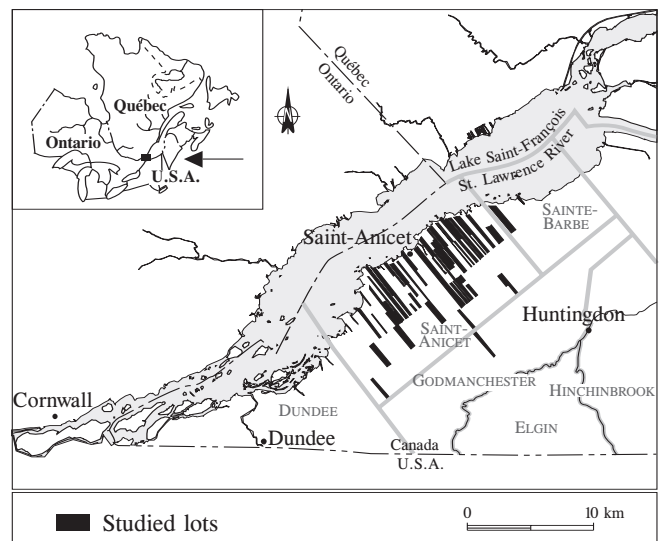


FIGURE 1. Haut Saint-Laurent region and location of the studied lots.

bitternut hickory (*Carya cordiformis*), and white ash (*Fraxinus americana*). Forests growing on poorly drained soils or open sites are mostly dominated by red ash (*Fraxinus pennsylvanica*), red maple (*Acer rubrum*), American elm (*Ulmus americana*), and eastern white cedar (*Thuja occidentalis*). White pine (*Pinus strobus*), red maple, trembling aspen (*Populus tremuloides*), and gray birch (*Betula populifolia*) are the main dominant species on sandy soils. Hardwood swamps, dominated by silver maple (*Acer saccharinum*), black ash (*Fraxinus nigra*), and red maple, and marshes occur along the shore of Lake St-François. Gray birch, red maple, or shrub communities dominate in peatlands.

Godmanchester Township, with its original boundaries, which included Saint-Anicet and Ste-Barbe until 1855, was first delimited and surveyed in 1785 and rapidly granted to officers and soldiers of the former Canadian Corps. However, few of them took possession of their land, and even 30 years later, Surveyor General J. Bouchette noted about Godmanchester that "only a small proportion is yet settled upon, and which is about the margin of the lake, where a few houses now and then present themselves, at considerable distances from each other; on the bank of the Chateauguay, also, there are a few settlements" (Bouchette, 1815). While the clearings were small, the main activity of these few settlers, and also of non-resident seasonal workers, was lumbering. The main targets were oaks and pines that were intended for the British market, especially from 1806 onwards, when the embargo imposed by Napoleon forced Great Britain to get its wood supply from its colonies. Potash production, using ash from cut and burned deciduous trees, was another lucrative activity having an impact on the forest (Booth, 1971). But it was not until the 1820s that extensive colonization of the township, strongly encouraged by the government, really started. Early settlements scattered in the area were often segregated according to the origin of settlers; hence, names such as "Canadian settlement" (*i.e.*, of French origin), "Irish settlement", "Scotch settlement", and "English and American settlement" on an 1831 map (Bouchette, 1831). With the progressive deforestation and depletion of merchantable timber, the economic contribution of forest exploitation rapidly decreased, being replaced by agriculture and cattle grazing. By 1888, agricultural development was such that the region was described as "one of the most cultivated of the Dominion" (*i.e.*, Canada with the boundaries of the time) (Sellar, 1888). Most of the remaining forests were woodlots kept for personal use of landowners (mainly for firewood) or forests located on soils that are unsuitable for agriculture. During the 20th century, more drainage and land reclamation provided new cultivated land, while a certain proportion of the old cultivated lands were abandoned and allowed to reforest, so the total proportion of forested land remained more or less constant. In recent years, the rising economic value of cultivated lands for corn and manure application have resulted in a renewed increase in deforestation.

Nearly two hundred years of severe deforestation, drainage of marshes and bogs for land reclamation, cattle grazing, and episodes of increase and decline in agricul-

ture have resulted in a complex vegetation mosaic (Bouchard & Domon, 1997). Today, the clay plains are used for agriculture, and the moraine deposits are colonized by fragmented forests that have experienced disturbances of various types and intensities.

NOTARY DEEDS

The notary deeds examined cover the period 1800-1880. Of the nearly 500,000 notary deeds considered, 448 involve one or more wood sales (Simard & Bouchard, 1996). For each wood sale, the names of the seller and buyer, the quantity of wood, the type (species), and its price were usually recorded. Most, but not all, of the French names for wood types used in the deeds allow a direct correspondence with a tree species (Bouchard *et al.*, 1989; Simard & Bouchard, 1996). "Orme" (elm) most likely refers to American elm, but may occasionally include the far less abundant red elm (*Ulmus rubra*). "Frêne" (ash) includes white ash, red ash, and, to a lesser extent, black ash. "Chêne" (oak) is probably always bur oak (*Quercus macrocarpa*), but red oak (*Quercus rubra*) may not entirely be ruled out in all cases. Red oak, however, does not usually establish well on calcareous soils with relatively high pH (Burns & Honkala, 1990) like the ones that predominate in our area (Timbal *et al.*, 1994). "Épinette" (spruce) is probably always black spruce (*Picea mariana*). "Épinette rouge" is the old vernacular name of tamarack (*Larix laricina*) and very likely refers to this species and not to red spruce (*Picea rubens*), of which "Épinette rouge" is now the modern French name.

Wood was sold in various forms: square logs, boards, shingles, cords (firewood), posts, staves, cribs, etc. For analysis, we converted each wood volume to cubic metres, using the appropriate formulae (Simard, 1995).

Although the identity of the seller is given in the notary deeds, neither his address nor the origin of the harvested wood is noted. Using other historical archives, we determined, whenever possible, the lot owned by the seller at the time of the transaction. We can reasonably assume that this lot is where the wood sold had been harvested. We eliminated from our study wood sales involving sellers whose identity was uncertain (*e.g.*, when two contemporary persons with the same name lived in the area) or who owned two or more distant lots. Also, two wood sales done at different times may have come from lots that were partially overlapping due to lot subdivision and sale. In these cases, only sales from the smallest lot were retained: the correspondence between the location of the harvesting and today's forest is more accurate on small lots than on larger ones. A few sellers made more than one wood sale from the same lot over time.

In all, 119 wood sales made on a total of 60 lots were retained for our study. The transactions were done between 1799 and 1870, most of them between 1820 and 1860 (Figure 2). The vast majority of the trees that were harvested during this period must have been established during presettlement times, so they provide an indication of the forest composition before the effect of human activities. While all the notary deeds from Godmanchester were examined, all wood sales come from lots located in

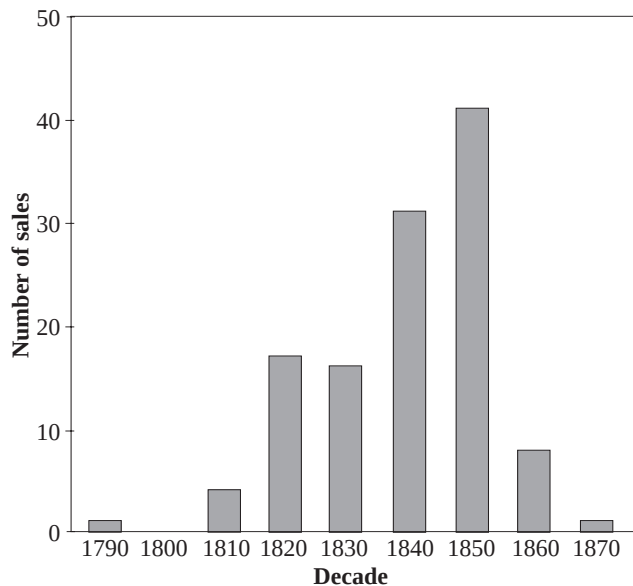


FIGURE 2. Number of wood sales considered in the study per decade.

the area formerly known as the “Canadian settlement”, which corresponds today to the village of St-Anicet and its vicinity (Figure 1). As the name implies, this area was settled by French Canadians, who made great use of the notary system in their transactions, unlike the American and British settlers who colonized other parts of the township. The average surface area of the studied lots at the time of the sales was 446 ha (min. 60 ha, max. 1,403 ha). Most lots were subsequently subdivided into smaller ones, often more than once, to arrive at the present allotment.

For a given location and seller, we determined the volume of wood sold for each tree species. All wood sales made by a given seller at different times (never more than a few years apart), and thus from trees presumably harvested at the same location, were grouped. In all, we determined the proportion of wood sold by species from 60 locations for which we could determine the exact boundaries today. These proportions are assumed to be a rough estimate of past forest composition at a given site. Only species with more than four mentions in the deeds were considered. Other than composition, factors that may have influenced the proportion of each species sold include wood value and uses, as well as management practices and personal preferences. Despite these possible sources of bias, the compositional patterns revealed by previous studies using notary deeds were generally found to be coherent and consistent with information obtained from other historical sources (Bouchard *et al.*, 1989; Simard & Bouchard, 1996). Nevertheless, to minimize errors, we chose to limit the interpretation of our results to the most significant and most consistent patterns.

DATA COLLECTION

Within the boundaries of each lot corresponding to a wood sale in the notary deeds, we determined the actual forest cover, when present. Each lot was visited, and the tree composition of the entire existing forested portion was sampled using a centred-point plots method. The number of plots depended on the size of the forested por-

tion of the lot, up to a maximum of 82 points for the largest one. Plots were regularly located within a forest. Their distance apart varied according to forest size, but was never less than 20 m in order to ensure that the same tree was not sampled twice. For each centred-point plot, we noted the species and diameter at breast height (DBH) of the nearest tree (DBH greater than 10 cm) in each of the four quadrants. We also verified the presence of trees from species that were formerly sold in the lot under consideration but that were not recorded in the centred-point samples with a meticulous exploration of the entire forested portion.

The relative importance of tree species was calculated for each lot in two ways: first, using all sampled species to determine present forest composition and, second, considering only species or species assemblages reported in the notary deeds to determine changes in composition. Most of the species present today but absent from the notary deeds either have no economic value or have never been common enough for significant exploitation. To facilitate comparisons between present tree composition and volumes sold from deeds, we based the relative importance of present-day species on the sum of DBH³. This cube transformation ensured that the units of both the past and the present measures of relative importance were the same. Since the present-day volumes were to be qualitatively compared to those from the notary deeds, which only represent a rough estimate of past composition, we found it unnecessary to use more accurate models of tree volumes for present-day composition.

Results

Of the 60 lots where wood sales were reported in notary deeds, 13 are entirely deforested today, and many others had only a small proportion of the lot still forested. The number of sampled trees ranged from 16 for the smallest forest to 280 for the largest one (average=87.1). The forested portions are most often found on a mesic, morainic portion of the lots, while the lowlands generally have been deforested for agricultural uses. Forests are usually young, and most show signs of uses and management, generally partial or severe cut, and past or present cattle grazing. Many woodlots are sugar maple stands. In fact, all sites considered, sugar maple is the dominant tree species in terms of relative density (17%); because of the larger average size of its trees, it is even more dominant in terms of volume (40%) (Table I). None of the maple groves showed signs of recent or older management for maple syrup. Many other woodlots are dominated by pioneer tree species, especially trembling aspen, red ash, and gray birch. For all sites combined, tree diversity is high, with 31 tree species noted, including one non-native species (Manitoba maple: *Acer negundo*) (Table I).

In all, 11 species or groups of species were sufficiently abundant in the notary deeds (appear in wood sales at least four times) to be considered (Table II). Species that were eliminated are paper birch (*Betula papyrifera*) and red maple/silver maple (“plaine” or soft maples). The largest volumes of wood sold were from sugar maple, American beech, and yellow birch, followed

TABLE I. Relative frequency and volume of tree species in the sampled plots. Species are in decreasing order of relative frequency.

English name	Tree species	Scientific name	Relative frequency (%)	Relative volume (%)
Sugar maple		<i>Acer saccharum</i>	17.4	40.3
Trembling aspen		<i>Populus tremuloides</i>	13.5	9.9
Red ash		<i>Fraxinus pennsylvanica</i>	10.6	5.6
Red maple		<i>Acer rubrum</i>	6.0	3.8
Hop-hornbeam		<i>Ostrya virginiana</i>	5.6	2.1
Bitternut hickory		<i>Carya cordiformis</i>	5.3	4.7
American elm		<i>Ulmus americana</i>	5.2	2.2
Eastern white cedar		<i>Thuja occidentalis</i>	5.1	2.2
Gray birch		<i>Betula populifolia</i>	4.0	1.3
American basswood		<i>Tilia americana</i>	3.6	3.2
White ash		<i>Fraxinus americana</i>	3.1	2.2
Black cherry		<i>Prunus serotina</i>	2.4	1.7
Silver maple		<i>Acer saccharinum</i>	1.8	3.7
Butternut		<i>Juglans cinerea</i>	1.7	2.6
White birch		<i>Betula papyrifera</i>	1.6	1.1
Manitoba maple		<i>Acer negundo</i>	1.5	0.3
Largetooth aspen		<i>Populus grandidentata</i>	1.4	1.1
Eastern hemlock		<i>Tsuga canadensis</i>	1.4	1.2
Eastern cottonwood		<i>Populus deltoides</i>	1.2	1.7
Bur oak		<i>Quercus macrocarpa</i>	1.1	3.7
Black ash		<i>Fraxinus nigra</i>	1.1	0.6
Willows		<i>Salix</i> spp.	0.9	0.5
Balsam poplar		<i>Populus balsamifera</i>	0.7	0.4
Pin cherry		<i>Prunus pensylvanica</i>	0.7	0.2
Yellow birch		<i>Betula alleghaniensis</i>	0.7	0.7
Balsam fir		<i>Abies balsamea</i>	0.5	0.2
White pine		<i>Pinus strobus</i>	0.5	0.7
American beech		<i>Fagus grandifolia</i>	0.4	0.8
Red elm		<i>Ulmus rubra</i>	0.3	0.3
Red oak		<i>Quercus rubra</i>	0.2	0.4
White spruce		<i>Picea glauca</i>	0.1	0.2

by white pine and hemlock. There were marked differences in terms of relative volume between the species reported in the notary deeds and those present in the forested portion today, when only the species sold are considered (Table II). Some species, such as sugar maple and the ashes, showed important increases, while others, such as beech and yellow birch, decreased sharply. On a site-by-site comparison, there were many instances where a species sold from a particular lot was completely absent as a tree from this same lot today (Table III). The nature of the differences in species composition may represent either a real change in time on a specific portion of the lot that is still forested or a change in the spatial extent of the forest cover: sold trees may have been harvested from a location on the lot that no longer supports a forest.

Of the 41 sites from which sugar maple was sold, 33 still support a forested portion today. Of these, 11 do not have sugar maple trees any more (Table III). These sites are mainly occupied by pioneer species such as trembling aspen and ashes, and one is dominated by eastern white cedar. Despite the absence of sugar maple at sites where it had been present, its overall percentage in terms of volume in today's forests is greater than its representation in the notary deeds (Table II). The situation is markedly different for beech and yellow birch, which were well represented in the notary deeds, but have a very low representation in today's forests, with 0.8% and 0.7% of total tree volume respectively (Table I) or 1.2% and 1.1% if only the species represented in the notary deeds are considered (Table II). Yellow birch is still present in only 5

TABLE II. Relative volume of tree species or group of species sold as reported in the notary deeds of the studied lots and contemporary relative volume of the same species (totalled over 100%) in the remaining forested portion of these lots.

	Wood sold in notary deeds Relative volume (%)	Present forest tree composition Relative volume (%)
Sugar maple	33.1	65.7
American beech	21.4	1.2
Yellow birch	18.7	1.1
Eastern hemlock	8.3	1.9
White pine	7.7	1.2
Tamarack	4.0	0.0
Black spruce	2.3	0.0
American elm, red elm	1.7	4.0
White ash, red ash, black ash	1.1	13.7
Eastern white cedar	1.0	3.6
Bur oak, red oak	0.8	7.5

of the 29 sites from which it was formerly sold where there is still a forest today, while beech is present in only 3 of the 27 sites still forested from which it was sold. In all these cases, their proportion in today's forest is lower than in the corresponding notary deeds.

Together, sugar maple, beech, and yellow birch represent 73.2% of the total volume sold, mostly for firewood (Table II). These species were also commonly sold together, which suggests that the sugar maple-beech-yellow birch forest type (Berglund, 1980) was well represented in the Haut-Saint-Laurent region. During the period covered by the notary deeds, the price of the three species

TABLE III. Absence of tree species on site where they were previously reported in the notary deeds.

	Number of sites where it was sold	Number of sites still supporting a forest cover	Number of forested sites where the species has not been noted
Sugar maple	41	33	11
Yellow birch	36	29	24
American beech	29	27	24
White ash, red ash, black ash	6	6	0
Tamarack	8	5	5
Black spruce	5	4	4
White pine	10	8	7
Bur oak, red oak	4	3	1
Eastern white cedar	13	9	4
Eastern hemlock	12	8	8
American elm, red elm	4	3	0

was the same (Simard & Bouchard, 1996), so their relative abundance in the notary deeds, when sold together, likely reflects their relative abundance in the forest. For the 19 sites where the three species were sold together, the average volume sold was roughly the same, *i.e.*, 34%, 27%, and 23% for sugar maple, beech, and yellow birch respectively (Table IV). Today, these species are absent from 6 of those sites, which are now dominated by pioneer species. In the 13 remaining sites, sugar maple is by far the dominant species overall, accounting for more than 25% by volume in all but two sites and reaching more than 50% in 5 sites. Beech and yellow birch were not noted in 7 of the 11 sites that can be considered as sugar maple groves. In fact, in all those sites, the relative abundance of beech exceeds 10% in only two cases (14% and 33%), while yellow birch never does (Table IV).

In the Haut-Saint-Laurent region, tamarack and black spruce are presumably associated with humid, bog-like conditions. Not only are they absent from the sites from which they were previously sold, but their presence has not been noted in any of the 60 studied sites (Table II).

The proportions of hemlock and white pine are lower in today's forest than in the notary deeds (Table II). Hemlock is absent from the 8 forested sites from which it had been previously harvested, and white pine is present in only one of the 8 sites from which a wood sale is reported (Table III).

Despite the occurrence of Dutch elm disease in the 20th century, the proportion of American elm in today's sites is slightly greater than its representation in the notary deeds (Table II). It is still present in the 3 forested sites from which it was previously sold (Table III). A similar trend is found in eastern white cedar, which was frequently sold, but in small volumes. This species shows a slight increase, from 1.0% in the notary deeds to 3.6% in today's forests (species sold only), and it is still present in 5 of the 9 forested sites from which it was previously sold. The stems of both of these species are generally of small size, which explains why elm and cedar each represents only 2.2% of the total tree volume (all species considered), while they represent 5.2% and 5.1% of the tree frequency, respectively.

In the studied sites, bur oak is represented by a few individuals of large size, often isolated in a pasture or accompanied by much smaller trees of other species. All species considered, it represents only 1.1% of the frequency, but 3.7% of the volume. Its relative dominance increased from 0.8% in the notary deeds to 7.5% in today's forests, when only the species sold are considered.

Ashes, as a group, comprise another taxon that is better represented today than in the notary deeds, going from 1.1% to 13.7%. Red ash is the most common of them today, with 10.6% of total frequency (5.6% of total volume). White ash comes second (frequency=3.1%; volume=2.2%), followed by black ash (frequency=1.1%; volume=0.6%). Although we cannot discriminate among species in the notary deeds, it is apparent that red ash, and maybe white ash, are more abundant in the studied sites today than previously, if their volume sold, as reported in the notary deeds, is representative of their former abundance in the forests.

Discussion

ORIGINAL VEGETATION ON LOWLANDS

In the Haut-Saint-Laurent region as a whole, and in St-Anicet in particular, the lowlands (mainly clay plain) largely dominate the landscape in terms of surface area. Yet, we know little of the original vegetation types in the lowlands because they were almost completely deforested for agricultural uses during the 19th century. Before it was drained for agricultural purposes, the clay plain had very humid soil conditions, with many swampy areas. Moreover, a large portion of what is now clay plain was originally a bog referred to as Tea Field Swamp (Jean & Bouchard, 1987). An 1863 map of the area shows that this bog covered the southeastern part of Godmanchester and a vast portion of the neighbouring township to the east (Bland, 1863). The major part of this bog originally had only a thin layer of organic material, facilitating peat moss scraping or burning to allow agriculture on the underlying mineral soil. Today, the remaining bog, which lies mostly east of the study area, has been reduced to two patches of 1,600 ha combined, compared to the 5,075 ha of the original Tea Field Swamp as shown on the 1863 map (Bouchard & Jean, 2001). It is likely that the size of the bog had already been reduced prior to 1863. Besides Tea Field Swamp, marshes, smaller bogs, or just very humid conditions probably were prevalent in a large part of what is now the lowlands. Sellar (1888) reports about the area around St-Anicet, "Owing to the country south of the lake being intersected by swamps the making of roads was beyond the ability of the settlers.... The record for the settlers [is] ... a contest with low-lying ground, hard to clear and harder to drain" This contest has long since been won, and today most of the clay plain is under cultivation.

Although collectively the studied lots comprise mostly lowlands, presumably of humid conditions, the largest proportion of trees sold by the early settlers were from species most commonly found under the more mesic conditions of the morainic ridges (sugar maple, beech, hem-

TABLE IV. Combination of sugar maple, yellow birch, and American beech in notary deeds and in present forests. Only sites where the three species were sold are considered. Each line corresponds to a site.

Site number	Year of sale	Proportion of each species in the sale as reported in notary deeds			Proportion of each species in the present forest cover		
		Maple	Birch	Beech	Maple	Birch	Beech
21	1843-46	0.41	0.27	0.33	0.96	0	0
75	1844-45	0.47	0.19	0.34	0.87	0	0
7	1827-32	0.21	0.37	0.17	0.81	0	0
67	1837-59	0.28	0.18	0.17	0.66	0	0
9	1817-23	0.52	0.15	0.15	0.59	0	0
77	1844-56	0.23	0.35	0.19	0.36	0	0.14
14	1799-1827	0.23	0.25	0.25	0.36	0	0
16	1833-41	0.71	0.07	0.13	0.36	0	0
76	1844	0.28	0.09	0.63	0.29	0	0.01
55	1841	0.08	0.46	0.46	0.27	0	0.32
37	1824-27	0.26	0.26	0.19	0.27	0.04	0
46	1846-51	0.36	0.05	0.05	0.09	0	0
71	1859	0.33	0.33	0.33	0.02	0.07	0.06
33	1839-42	0.29	0.14	0.57	0	0	0
20	1827-41	0.30	0.20	0.48	0	0	0
59	1851	0.26	0.26	0.48	0	0	0
62	1853-55	0.50	0.33	0.17	0	0	0
63	1843	0.50	0.21	0.44	0	0	0
70	1851-59	0.18	0.11	0.15	0	0	0
Average		0.34	0.23	0.27	0.47*	0.01*	0.04*

*Only sites where at least one of the three species is present are considered for the average.

lock, and to a lesser extent yellow birch). The apparent discrepancy between the predominance of lowlands in the landscape and the low representation in the notary deeds of tree species that prefer more humid conditions suggests that precolonial forests on the lowlands were sparse or patchy or contained tree species of little stature or commercial value.

Tamarack and black spruce may have been somewhat more abundant in swamps, but they were of little economic value. These species are no longer present where they were harvested to be sold, or in any other sampled lots. In fact, both these species have been almost extirpated from the area. In the Haut-Saint-Laurent region as a whole, tamarack is sparse, and the species is mostly confined to a small protected coastal swamp (Jean & Bouchard, 1991). Black spruce was once abundant in the Tea Field Swamp, but a recent extensive survey of the remaining portion of the swamp failed to find a single individual (Jean & Bouchard, 1987).

In theory, red maple and silver maple could have colonized the clay or loamy soils with imperfect drainage that were subsequently deforested. Today, red maple is abundant in the Haut-Saint-Laurent, especially in disturbed forests on sandy soils. However, the very low representation of "soft maple" in the notary deeds despite their possible commercial value as firewood suggests that they were poorly represented in the lowlands.

Primarily used for posts, eastern white cedar does appear in the notary deeds and was sold mainly as poles or assembled in rafts and floated to its destination. However, its price was low, so its representation in the notary deeds may underestimate its true original abundance. It may have been common in some parts of the lowlands, because the species is known to withstand conditions including humid depressions and organic soils (Johnston, 1990). Today, it is still abundant, notably

because it benefited from cattle browsing of deciduous competitors, and has successfully invaded abandoned pastures of any soil conditions, including mesic sites of morainic origin (de Blois & Bouchard, 1995).

American elm may have been more abundant on the original lowland than suggested by the notary deeds. Because it produces large quantities of ash, a greater proportion of elm may have been burned for potash production rather than sold as wood in comparison with other tree species (Booth, 1971).

Two species, oak and white pine, were extensively harvested before settlement took place. Of these two, bur oak was mostly found on the clay plain, especially along the waterways. Its early exploitation was such that it may well have been significantly reduced in number by the time colonization started, *i.e.*, the time covered by our study. In 1820, J. Bouchette had already noticed the reduction in trees of economic value, noting the "cutting and rafting of timber, a concern that has been carried on extensively for many years, and in consequences of which timber of large size, principally oak and pine, has become rather scarce in the vicinity of the lake, rivers and creeks within this township" (cited in Sellar, 1888).

ORIGINAL VEGETATION ON MESIC SITES: MODIFICATION OF THE ORIGINAL SUGAR MAPLE-BEECH-YELLOW BIRCH FOREST

In the Haut-Saint-Laurent region, mesic sites occur mainly on morainic deposits, which represent approximately 22% of the combined territory of our studied lots. Although a few of these rocky sites may have been cultivated in the past, today's requirements for farm equipment render them non-arable. A large proportion of them are thus occupied by forests of various ages and degrees of disturbance. Many sites are dominated by early successional species, an indication of a recent history of severe cut or field abandonment. Others are dominated by late successional species, especially sugar maple, but partial

cuts and other type of management in several of these forests have created openings that allow shade-intolerant or mid-tolerant species to persist. This situation is typical of the Haut-Saint-Laurent as a whole (Brisson, Bergeron & Bouchard, 1988; Bergeron *et al.*, 1988; Meilleur, Bouchard & Bergeron, 1994). While current composition may depend on the severity or nature of past disturbance, studies on succession in this area show that these stands all converge toward a stable maple-hickory community type, where sugar maple, largely dominant, is accompanied by various species such as bitternut hickory, American basswood and hornbeam (Brisson, Bergeron & Bouchard, 1988; Bergeron *et al.*, 1988). In the Haut-Saint-Laurent, many mesic forests of natural appearance and with little sign of recent disturbance do correspond to this community type, including some of the forest sites on morainic ridges in the present study. Yet, there seems to be a significant difference between this apparent equilibrium state and the original composition on mesic sites in the studied area. Our results suggest that beech and yellow birch were strongly co-dominant with sugar maple and that the presettlement forest of mesic sites in the Haut-Saint-Laurent region of Québec corresponds better to the sugar maple-beech-yellow birch forest type (Berglund, 1980). The most important change since European settlement on mesic sites thus appears to be a loss or decrease in the abundance of beech and yellow birch, to the benefit of sugar maple and maybe other tree species. In present-day Haut-Saint-Laurent, beech presence and abundance are irregular and patchy. It is occasionally present in association with maple or hemlock on well- to moderately drained sites, but it is absent from a large proportion of forest stands where edaphic conditions otherwise appear to meet its ecological requirements (Bergeron *et al.*, 1988; Meilleur, Bouchard & Bergeron, 1994).

While wood sales recorded in notary deeds may be the strongest evidence of a greater presettlement abundance of beech in the Haut-Saint-Laurent, there are other indications in support of a larger representation of this species at the beginning of European occupation. Surveyor General J. Bouchette wrote in 1815 about neighbouring Hinchinbrooke that "The large knolls, or rising ground, are thickly clothed with good timber, as beech, maple, birch, pine, oak, butternut, and basswood" (Bouchette, 1815). A few years later, he noted about the same area that "The timber is of superior quality, namely beech, maple, elm, basswood and pine", and about nearby Hemminford that "it is chiefly timbered with beech, birch, maple, elm, pine and basswood" (cited in Sellar, 1888). Note the order in which the species are mentioned, which may well represent a rough ranking of decreasing abundance. Sellar (1888), reporting the words of an early settler of St. Anicet who had poor knowledge of wood uses at his arrival, wrote, "Because it was so easy to chop and split, we preferred basswood for firewood, leaving the beech and maple, and had poor fires in consequences."

Another indication of the past abundance of beech is the existence of a unique old-growth beech-maple forest on a morainic ridge of the Haut-Saint-Laurent region. This 11-hectare forest, the Boisé-des-Muir Ecological

Reserve, has been exceptionally well preserved and never suffered severe partial cuts or other forms of exploitation (Brisson, Bergeron & Bouchard, 1992; Brisson *et al.*, 1994). When this forest was first studied in the 1980s, its difference in composition from the hypothesized maple-hickory climax type for the region, and notably the abundance of beech, appeared surprising and raised the possibility that the Boisé-des-Muir, although an old-growth forest, might not be entirely representative of the average mesic pre-settlement forest. The present findings from wood sales in the notary deeds and indications from other sources now suggest that the Boisé-des-Muir is just additional evidence of the past abundance of beech and its subsequent decline due to human activities.

In eastern North America, a decline in beech abundance may already have been under way before European arrival, attributable either to climatic change or to Native American activity (Clark & Royall, 1995; Fuller *et al.*, 1998). A much accelerated and general decrease in beech since European settlement, however, has been noted repeatedly across most of its range and, in fact, is often described as one of the most dramatic changes in tree species composition (Cogbill, 2000). For instance, south of Québec, in northern New England, beech is always the most abundant tree in survey records, yet its current relative dominance is only from 10% to 20% of its presettlement abundance (Siccama, 1971; Cogbill, 2000). Other studies documenting a decline in beech abundance since European colonization include Russell *et al.* (1993), Foster (1992), McIntosh (1972), Shanks (1953), and Ward (1956). Only at the western edge of its range does beech show a sharp increase in its dominance (Woods, 2000).

The increase of early-successional species at the expense of late-successional species due to human activities is one obvious cause of the decline of beech in most of its range. Other factors often mentioned as responsible are lack of adequate regeneration, beech bark disease, or over-utilization for wood, but none of these factors satisfactorily explains the extent of the decline compared to the presettlement abundance of beech (Siccama, 1971; Cogbill, 2000). Based on paleoecological records, Fuller *et al.* (1998) suggest that human activities may simply have accelerated a gradual decline in beech already underway before European settlement. In our study, we found no beech trees in some of the late-successional forest stands that once provided beech for firewood in the early 19th century, and most of these stands now appear to be devoid of any beech seedlings or saplings. Landscape fragmentation has isolated some of these stands from nearby sources of propagules. Even where beech is present in the canopy, species maintenance relies heavily on root sprouting. For example, in the Boisé-des-Muir Ecological Reserve, beech seedlings have low survival rates, and most of the seedlings and saplings above 30 cm high are from sprout origin (Beaudet *et al.*, 1999). South of the study area, at Lake George, New York, Nicholson, Scott, and Breisch (1979) also noted that "beech is not reproducing well, and many of the smaller stems that appear to be saplings are in fact of sprout origin." While beech sprouts allow self-maintenance and even a progressive increase of beech dominance over time (Brisson *et*

al., 1994), the species' limited seed dispersal and low seedling establishment may not allow beech re-establishment where it has been removed.

Compared with beech, yellow birch has a wide tolerance regarding moisture conditions (Erdmann, 1990). In the Haut-Saint-Laurent, the fact that yellow birch was most commonly sold with sugar maple and beech in approximately equal amounts suggests that, like beech, it was an important component of the mesic forests and that it subsequently decreased in abundance and disappeared from several maple stands. However, some of this yellow birch may have been cut on the lowland portion of the territory that is no longer forested. Today, yellow birch is uncommon in mesic forests of the Haut-Saint-Laurent, but it is still frequently found in more humid forests, both young and old, although generally in low abundance (Bergeron, Bouchard & Leduc, 1988; Meilleur, Bouchard & Bergeron, 1994). Yellow birch is known to grow best on well- to moderately well-drained sites, where it usually regenerates on mossy logs, decaying wood, and rotten stumps (Erdmann, 1990). Hardwood leaf litter is detrimental to its survival elsewhere because drying of the litter during the growing season kills most seedlings. By comparison, in less well-drained soils, sufficient moisture may remain in the leaf litter, allowing establishment, although subsequent growth is poor (Erdmann, 1990). In the Haut-Saint-Laurent, it is thus possible that forest management, which often results in fewer decaying trees and logs, has been more detrimental to yellow birch in mesic forests than in more humid sites.

In the Haut-Saint-Laurent, sugar maple is the most important component of forests on well- to moderately well-drained sites, and it is considered the dominant species of the hypothesized equilibrium state (Bergeron, Bouchard & Leduc, 1988). Yet, while sugar maple is considered a late-successional species, it may still benefit from human disturbances (Siccama, 1971; Nicholson, Scott & Breisch, 1979). It is often abundant in the understory of young secondary forests originating after clear-cuts or abandoned fields dominated in the canopy by pioneer tree species (Brisson, Bergeron & Bouchard, 1988). The large number of wind-dispersed seeds, high establishment rate, and shade tolerance of sugar maple all contribute to its resilience. Compared to other late-successional species, it has not suffered as large a decrease in terms of relative abundance since European colonization, and it has even shown an increase in some areas (Siccama, 1971; McIntosh, 1972). In the Haut-Saint-Laurent, the elimination of beech also may have benefited sugar maple. In the Boisé-des-Muir old-growth forest, maple is not reproducing well despite the abundance of seed-producing trees, apparently because of the deep shade of beech foliage (Beaudet *et al.*, 1999). Nicholson, Scott, and Breisch (1979) also noted that maple was barely reproducing in old-growth mesic stands dominated by beech or hemlock.

One of our sites is now occupied by a nearly pure eastern white cedar stand on a morainic ridge. Across its range, this species is known to have a wide tolerance to drainage conditions, from very wet soils with seasonal flood to very dry rock outcrops. However, while it grows well in mesic conditions, it is apparently much less com-

petitive than many deciduous species. In the Haut-Saint-Laurent region, it is therefore believed that originally, the species was confined mostly to the poorly drained lowlands. Its current abundance and occasional dominance on mesic sites is a recent phenomenon caused by cattle grazing followed by pasture abandonment (de Blois & Bouchard, 1995).

LONG-TERM EFFECT OF HUMAN ACTIVITIES ON VEGETATION DYNAMICS

Due to possible errors and biases, caution is always necessary when interpreting information from archival documents for presettlement forest reconstruction, even with the commonly used approach of witness trees from land surveys (Bourdo, 1956; Whitney, 1994). In the case of wood sales reported in notary deeds, sources of bias include wood commercial value and personal preferences (Simard & Bouchard, 1996). While the overall change in the relative abundance of species on a site revealed by this approach remains a broad approximation, the most notable and solid pattern that it uncovers is the common elimination of tree species from the canopy on a site-by-site basis. American beech, yellow birch, spruce, and tamarack are among those that repeatedly show this pattern of local elimination. The frequent lack of regeneration of these same species in the understory of many of these lots suggests that there is no tendency toward an eventual return to their original abundance. For some species, a return may not even be possible considering the profound and irreversible modifications that have occurred, like changes in the drainage system, peat elimination, or other soil modifications for agricultural purposes. While global climate change, accidental introduction of exotic tree diseases, or altered fire regimes are often mentioned as possible anthropogenic causes of changes in species composition (Russell *et al.*, 1993), other human-caused factors that have attracted little attention may have a significant impact on tree establishment and forest dynamics. For example, the introduction of earthworms of the Lumbricidae family, previously nonexistent in eastern Canada, may have had a significant impact on litter decomposition, soil structure, and nutrient transformation (Langmaid, 1964; Hendrix, 1995) and, consequently, plant germination and establishment. Also, it is very likely that the extinction of the passenger pigeon (usually attributed, partly or entirely, to overhunting), once considered the most abundant bird species in eastern North America, had an impact on beech regeneration or dispersion, positively or negatively, considering that beech nut was its main source of food (Shorger, 1955).

Forest cover in the Haut-Saint-Laurent region has been greatly reduced since European settlement. Tree composition of the forested portion remaining today is very different from the precolonial forest, and our results show that the difference is not only one of early- versus late-successional stage. Presettlement forest composition appears to be different from the steady states toward which succession apparently converges in the long run. In this agro-forested landscape, there is no indication of an eventual return toward the original forest composition, nor should there necessarily be one considering the extent

and the irreversible nature of some of the ecosystem modifications caused by human activities.

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